Trustworthy Multi-Agent ML via Cryptographically Grounded Discovery Tasks

A Testbed with Provable Ground Truth and Exact Information Leakage

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Motivation

Trust in multi-agent ML pipelines is normally asserted, not measured. When a system of agents proposes a conclusion, we lack the tools to say exactly how much a compromised member could damage the outcome, how many bits an untrusted channel leaked, or whether a verifier agent adds real assurance or merely a second opinion. Cryptography has precise language for exactly these questions — information leakage in bits, prover/verifier soundness, Byzantine fault tolerance under a threshold of corrupted parties — but that language requires a setting with known ground truth to instantiate, and most ML settings do not have one.

Evaluation of LLM-based agents on scientific discovery has a matching gap: scoring a discovered law presupposes knowledge of the true law, which is unavailable for the problems of real interest. Current discovery benchmarks substitute LLM judges or rubric matching, inheriting the reliability problems of the systems under evaluation. A second confound is contamination: when a system states a correct law, standard evaluations cannot distinguish derivation from memorization.

This work addresses both gaps with one construction: tasks whose generative law is a provable theorem, drawn from arithmetic statistics with a direct cryptographic reading. Because the law is known exactly, we can compute the Bayes-optimal ceiling, the exact information content of every observable in bits, and a machine-checkable answer key — turning "is this multi-agent pipeline trustworthy" from a qualitative judgment into a set of measured quantities.

Task family

Let $K/\mathbb{Q}$ be a Galois number field with $G = \mathrm{Gal}(K/\mathbb{Q})$, and let $\zeta_K(s) = \sum_{n\geq 1} a_n(K)\, n^{-s}$ be its Dedekind zeta function, so that $a_n(K)$ counts ideals of norm n. For a rational prime p with decomposition type (e, f, g) in K (ramification index, inertia degree, and number of primes above p, with $efg = [K{:}\mathbb{Q}]$), the p-Euler factor is $(1 - p^{-fs})^{-g}$, whence

$$a_{p^k}(K) \;=\; \binom{g + k/f - 1}{g - 1} \text{ if } f \mid k, \qquad a_{p^k}(K) = 0 \text{ otherwise,}$$

and a_n is determined multiplicatively. The inertia degree f at an unramified prime is the order of its Frobenius class, which by Chebotarev is equidistributed with respect to the counting measure on G. Consequently the isomorphism type of G leaves provable statistical fingerprints in the coefficient vector $(a_1, \ldots, a_N)$: only element orders of G occur as inertia degrees, with prescribed densities. A representative criterion: for $[K{:}\mathbb{Q}] = \ell^2$, the extension is cyclic iff $a_{p^\ell}(K) = 0$ for some prime p; only C_{ℓ^2} admits inert primes, and in the noncyclic case a_{p^ℓ} is provably nonvanishing at every prime, ramified or not [1]. Each benchmark task is a classification problem: given $(a_1, \ldots, a_N)$ for a field drawn from this generative model, identify G among the groups of the given order, and state the distinguishing criterion. The setting follows Lee and Lee [1], who classified such G with interpretable models and promoted the learned decision logic to new theorems.

The benchmark distribution is the exact Chebotarev model: independent Frobenius classes at each prime, i.e. the idealized unramified statistics. This idealization is validated against genuine arithmetic: on number fields drawn

from the LMFDB (with zeta data computed from defining polynomials via Dedekind's theorem), the proven quartic criterion classifies at accuracy 1.000, as the theorem guarantees.

Exact ground truth

Because the generative model is known in closed form, each task ships with quantities that are typically unobtainable:

- **Bayes-optimal accuracy.** The class-conditional law of the observable prime signatures is an explicit categorical distribution per prime, so the Bayes risk is estimable to $\pm 5 \times 10^{-4}$ by direct signature sampling. Agent performance is reported as regret against this ceiling; the ceiling itself calibrates task difficulty (from 0.9998 down to 0.84 across the suite).
- **Exact attribution ground truth.** The mutual information $I(a_n; G)$ is computable in closed form for every feature via the independence of prime-power factors. A system's stated feature attributions are scored against this ranking (rank correlation, precision at k), which exposes cases where predictive accuracy and explanation fidelity decouple; we observe decision trees within 0.3% of the ceiling whose importances correlate weakly with the true ranking.
- **Reference theorems.** The correct explanation for each task is a proven statement (e.g. the inert-prime criterion above), serving as the answer key for conjecture grading.

A natural hardening step is formal verification of the answer keys. The discrete content of each criterion (the element-order multisets of the candidate groups, the vanishing pattern of the Euler coefficients as a function of the inertia degree, and the resulting exclusion arguments) is directly formalizable in Lean 4 against Mathlib, making the ground truth machine-checked rather than merely peer-reviewed. The analytic inputs (Euler products, the Chebotarev density theorem) would enter as stated hypotheses pending their availability in Mathlib. We view machine-checked answer keys as the natural endpoint of the benchmark's premise: an evaluation is only as trustworthy as its ground truth.

Cryptographic grounding

The exact-ground-truth machinery above is not merely convenient for grading; each piece has a direct cryptographic reading, which is what makes the testbed suitable for trustworthy-ML questions rather than capability questions alone.

Mutual information as leakage. $I(a_n; G)$ is, literally, the number of bits an observable coefficient leaks about a secret (the Galois group). Scoring a system's stated feature attributions against this exact quantity is the same primitive as side-channel leakage assessment: does the adversary's (here, the agent's) inferred importance ranking match the true leakage profile, or does it over-attend to non-leaking observables while missing the real channel? Because the leakage is computed in closed form rather than estimated from traces, this gives a noise-free testbed for leakage-detection methodology itself.

Analyst/critic as prover/verifier. The two-agent pipeline (Section on directions, below) is structurally an interactive proof: an analyst (prover) claims a rule; a critic (verifier) must accept or reject using only tool queries against the data, without an a priori trusted computation of the answer. Soundness (the verifier rejects false claims) and completeness (the verifier accepts true claims) are directly measurable, because the true claim is known to us even though it is not given to the verifier.

Poisoning as a chosen-distribution attack with exact damage. Because the generative model is closed form, a poisoning attack on the training distribution or on an agent's tool channel has a computable effect on the Bayes ceiling: the information destroyed by the attack is a bit quantity, and the performance actually lost by the pipeline is a separate, comparable bit quantity. This is the natural analogue of measuring an attack's advantage in a cryptographic security game, applied to a data-poisoning rather than a cipher-breaking adversary.

Byzantine tolerance as a threshold property. With k agents and t compromised, standard secure multi-party protocols guarantee correctness once t is below a known threshold. The same question is open and largely unmeasured for multi-agent *epistemic* pipelines: how does discovery accuracy degrade as a function of t, the

subtlety of the compromise, and the verification topology (single critic, quorum, adversarial critic)? Exact regret makes this a plottable curve rather than an anecdote.

Evaluation protocols and contamination design

Systems are evaluated under three protocols: *rule induction* (data and mathematical context provided; the system must return an executable decision rule and a stated conjecture, with the rule scored on held-out fields), *derivation* (mathematical context only, no data; tests whether the criterion can be derived rather than fit), and *direct classification* (few-shot in-context prediction). The harness is model-agnostic: any prompt-to-text interface, including a multi-agent system behind a single entry point, plugs in unchanged; alternatively an agent may be given the dataset files and evaluated on its returned artifacts. Sandboxed execution and oracle/majority controls validate the pipeline independently of any model.

The generator requires only the element-order multiset of a candidate group, so the suite includes tasks for degrees 14, 21, and 25 whose distinguishing criteria follow from the same theory but, to our knowledge, appear in no publication. Matched comparisons between published and held-out tasks convert contamination from a hypothesis into a measurement.

Preliminary findings

A preliminary single-seed run of Claude Sonnet 4.5 exhibits the separations the design targets. Under rule induction the model attains near-ceiling regret on several tasks, including held-out degree 14 (regret 0.007). Under derivation it collapses to majority-class performance on most tasks, with one exception: the degree-9 task, whose criterion is published, where it achieves regret 0.004. This asymmetry is consistent with memorization of the published criterion; the decisive comparison is against the structurally matched held-out degree-25 task (both are the ℓ^2 family), which is part of the full multi-seed evaluation in progress. Notably, the model also scored 0.958 on the held-out degree-14 derivation, where no published criterion exists; whether its stated conjecture constitutes a correct derivation of the inert-prime argument is exactly what the conjecture-grading protocol assesses, and the harness now records all model outputs for this analysis.

Implemented instrument and current experiments

A first instrument for the trust questions above is implemented: a tool-using multi-agent pipeline (Strands agents on Amazon Bedrock) in which an analyst agent explores the data through tools (inspecting per-class coefficient distributions, testing candidate rules against a validation split) and proposes a rule and conjecture, and an independent critic agent verifies or revises it using the same tools. Two experiment families are in progress, both scored as regret against the same computable ceiling:

Agent-level compromise. Four conditions — analyst alone, analyst with critic, a simulated compromised analyst (instructed to produce a plausible but subtly flawed rule), and the compromised analyst with critic — make the value of verification and the harm of a compromised member directly comparable on the same tasks.

Tool-channel compromise. Holding the agent prompts honest and identical across conditions, we instead corrupt the environment: the observation tool reports falsified per-class distributions, the validation tool inflates reported accuracy and hides counterexamples, tool output carries injected operator-style instructions, or all three simultaneously with no honest channel remaining. An initial run surfaced a design lesson directly relevant to leakage assessment: when the task brief itself contains the group-theoretic facts needed to derive the rule, that channel is redundant and an attack on the tool channel alone cannot succeed, since the agent can bypass poisoned tools entirely. Measuring genuine attack harm requires withholding that redundant channel so the tool output is load-bearing — itself an instance of the leakage question above, since it is asking which channel actually carries the exploitable information.

These experiments are complementary to the APWA evaluation agenda and, we believe, a natural point of contact between multi-agent ML evaluation and the security group's expertise in cryptographic and adversarial analysis.

Code, benchmark suite (18 configurations, Hugging Face format), and harness: `https://github.com/mmjerge/galois-ml`.

References

[1] K.-H. Lee and S. Lee, *Machines Learn Number Fields, But How? The Case of Galois Groups*, arXiv:2508.06670 [math.NT], 2025. https://arxiv.org/abs/2508.06670